FAMILY-LEVEL COMPARISON

Kraken2/*KrakenUniq*

Do two metagenomic classifiers tell us the same biological story across 250 ancient DNA samples?
DNA samples?

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DATABASES
GTDB · MicrobialNT

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Two classifiers, one set of samples.

ANCIENT DNA SAMPLES

250

classified by **two tools** *at family level level*

Are Kraken2 and KrakenUniq giving us the *same biological picture?*

Five steps from raw reads to a comparison matrix matrix

STEP 1

Pre-processing

FastQC, Cutadapt, paired-end merging: same input for both classifiers.

250 cleaned FASTQs

STEP 2A

Kraken2 · GTDB

SLURM job array on Dardel.
Bacteria + archaea.

250 reports

STEP 2B

KrakenUniq · MicrobialNT

SLURM job array on Dardel.
Eukaryotes, fungi, viruses.

250 reports

STEP 3

Family matrices

Family rows extracted, *samples × families* matrices built on Dardel.

2 matrices · TSV

STEP 4

R analysis

Cleaning · per-family Spearman · Bray-Curtis → PCoA → Procrustes.

Today's talk

TAKE-HOME HOME

Same input, two parallel pipelines, two matrices

Two matrices and an early sanity check

FIRST EXPLORATORY PASS

Both tools see the same dominant biology

biology

Top family is ***Burkholderiaceae*** in both. The **same sample is highest**, the **same sample is lowest**, in both tools

HIGHEST BACTERIAL FRACTION

P35761_1115*

51.6% / 22.7%

LOWEST BACTERIAL FRACTION

P32015_1105

0.84% / 0.30%

* Note: some samples are missing from the *metadataGOG* file

OUTPUT OF THE PIPELINE

Two *samples × families* matrices

One row per sample, one column per family, percent abundance in each cell

Kraken2

GTDB · bacteria + archaea

archaea

249 × 5460

5460

KrakenUniq

MicrobialNT · incl. eukaryotes

eukaryotes

247 × 1662

Four choices that make the comparison fair.

Sample intersection (not union)

01

Used the 246 samples present in *both* tools.
Avoids zero-padding artifacts that bias multivariate analysis.

Removed Hominidae

02

Kraken2/GTDB has no eukaryotes.
Hominidae was 43% of KrakenUniq reads: keeping it would mask the microbial signal.

Excluded P34558_1102

03

Zero microbial reads in KrakenUniq after Hominidae removal: biologically interesting interesting (mostly human DNA), but unusable here.

Row-normalization to relative abundance

04

Each sample rescaled to `rowSum = 1` before Bray-Curtis.

FINAL ANALYSIS DATASET

245 samples × 362 shared families families

At the community level, the two tools *agree*

r = 0.945

SPEARMAN · PC1 VS PC1 · P < 0.001

0.945

PROCRUSTES (FULL BIVARIATE ALIGNMENT)

r = 0.90 · p < 0.001

- The dominant biological gradient is the same in both ordinations.
- Samples extreme in Kraken2 are also extreme in KrakenUniq.

KrakenUniq.

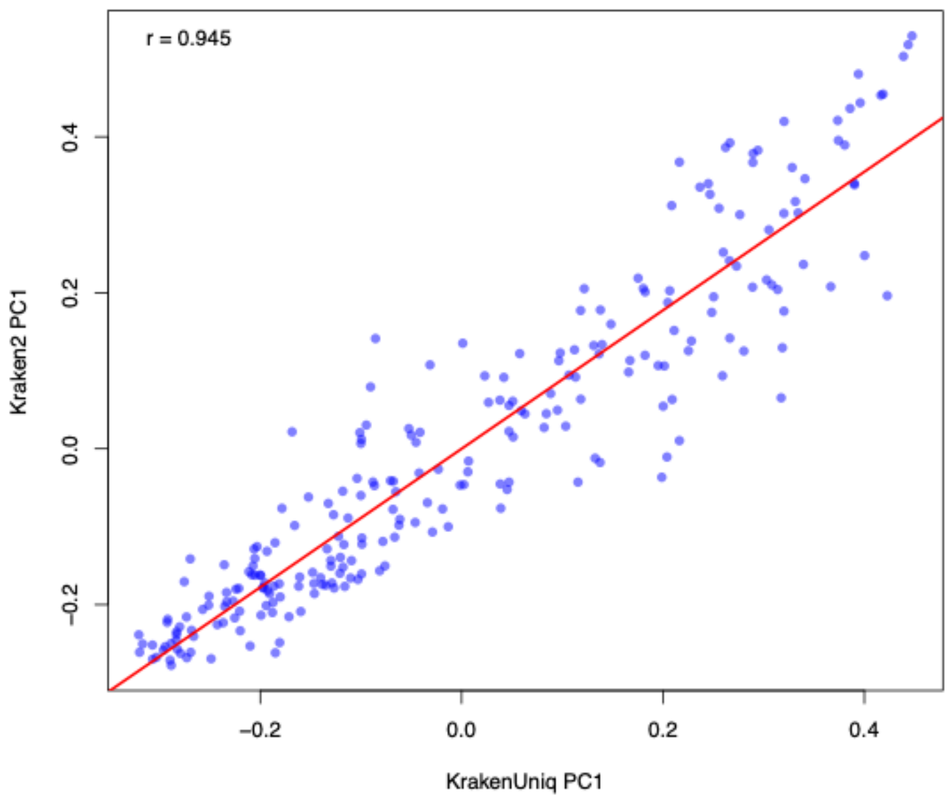


Fig. 1 · KrakenUniq PC1 (x) vs Kraken2 PC1 (y) · 245 samples · red line = 1:1

At the family level, agreement is concentrated in a few taxa

0.28

MEDIAN SPEARMAN P
ACROSS 201 VALID FAMILIES

19 FAMILIES WITH P > 0.8

Burkholderiaceae · Gemmatimonadaceae · Rhodanobacteraceae · Rhodanobacteraceae · Steroidobacteraceae · Micromonosporaceae Micromonosporaceae and others;
abundant environmental taxa.

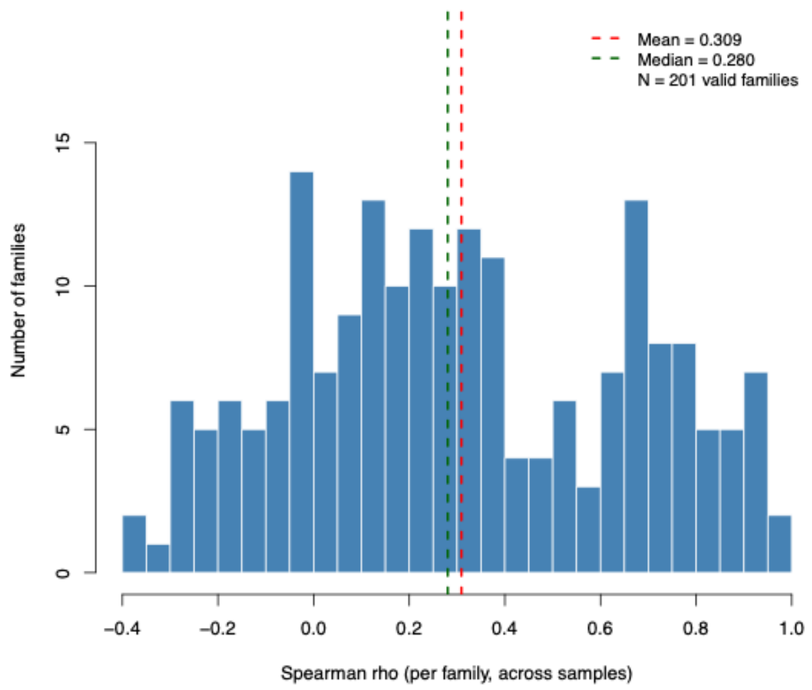
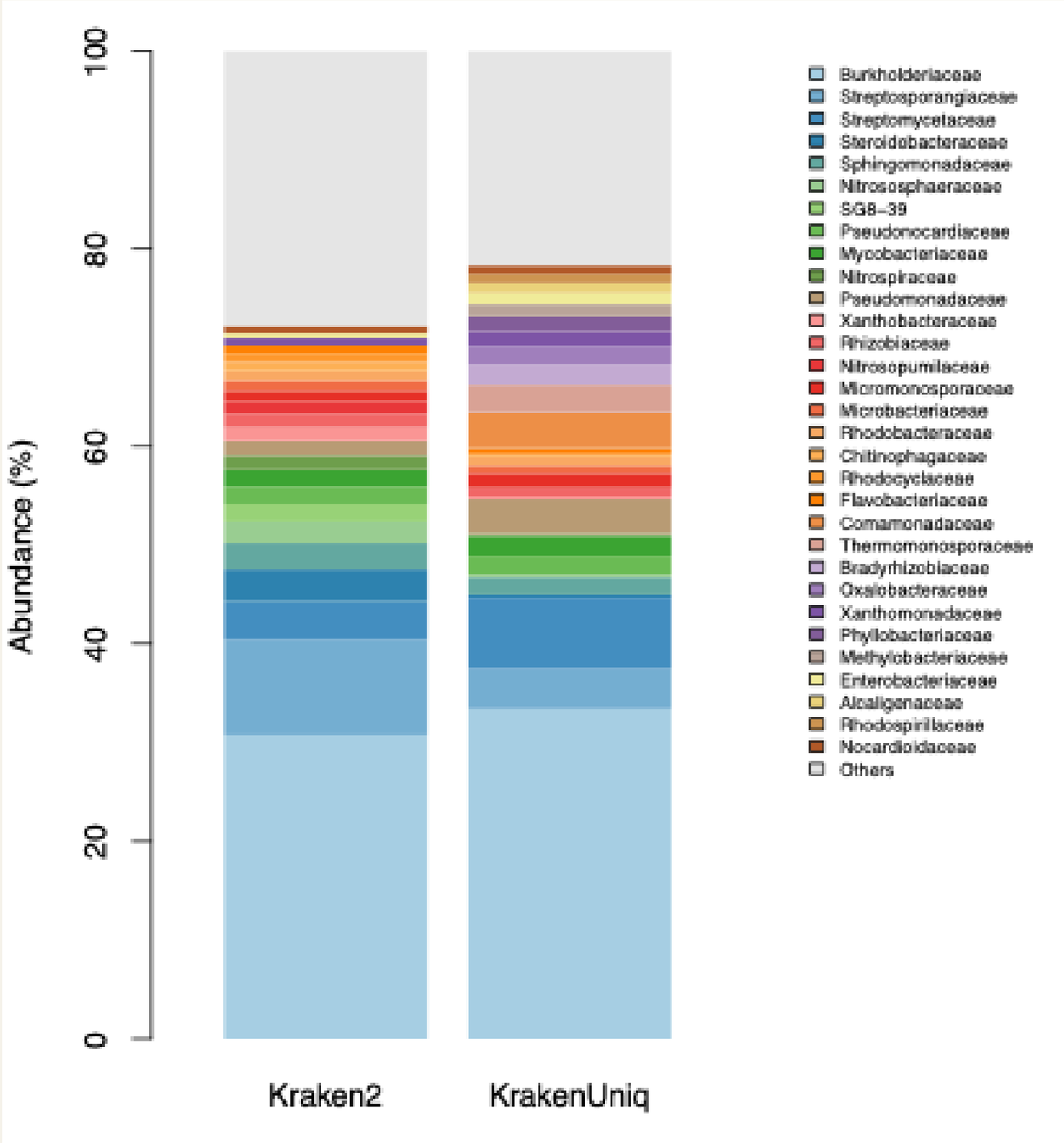


Fig. 2 · Per-family Spearman ρ across 201 valid families. How to read it → Bars on the right (ρ > 0.6) are families where both tools agree clearly. The disagreement sits on the left and middle — concentrated on rare families with sparse counts.


Abundance (%)
0
20
40
60
80
100
Kraken2
KrakenUniq
Burkholderiaceae
Streptosporangiaceae
Streptomycetaceae
Steroidobacteraceae
Sphingomonadaceae
Nitrososphaeraceae
SG8−39
Pseudonocardiaceae
Mycobacteriaceae
Nitrospiraceae
Pseudomonadaceae
Xanthobacteraceae
Rhizobiaceae
Nitrosopumilaceae
Micromonosporaceae
Microbacteriaceae
Rhodobacteraceae
Chitinophagaceae
Rhodocyclaceae
Flavobacteriaceae
Comamonadaceae
Thermomonosporaceae
Bradyrhizobiaceae
Oxalobacteraceae
Xanthomonadaceae
Phyllobacteriaceae
Methylobacteriaceae
Enterobacteriaceae
Alcaligenaceae
Rhodospirillaceae
Nocardioidaceae
Others

Differences

Kraken2-only family names

5,098

family names only in GTDB - mostly alphanumeric placeholders for unnamed clades

KrakenUniq-only family names

1,299

family names only in MicrobialNT - mostly eukaryotes, fungi, viruses not covered by GTDB

Shared by exact name

3622

family names shared by both databases - the only ones directly comparable in this analysis

What I *found.*

01

GLOBALLY

1 Same biological *story*

Community structure is preserved across tools
PC1 r = 0.945 , Procrustes r = 0.90.

02

LOCALLY

2 Limited by *naming*

Per-family agreement is moderate (median ρ = 0.28),
but the disagreement is mostly database labels, not biology.

naming

OPEN QUESTION · NEXT STEPS

STEPS

Which path next?

OPTION A

A

Search the existing reports for *Yersinia pestis* & *Tannerella forsythia*

forsythia

OPTION B

B

First aggregate to *Phylum* level: check whether per-family disagreement was just a resolution issue

resolution issue